Lattice Signature with Efficient Protocols

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1 Introduction

Digital signatures are fundamental cryptographic primitives, allowing a user to authenticate a message and guarantee its integrity. For decades, most practical signature schemes have relied on classical hardness assumptions such as integer factorization or discrete logarithms. However, these assumptions are known to be broken by quantum algorithms, which motivates the search for alternative constructions secure in the post-quantum setting. Lattice-based cryptography has emerged as one of the most promising frameworks for post-quantum security, as its underlying problems are believed to remain hard even for quantum computers. Beyond security, modern applications often require signatures to interact efficiently with zero-knowledge proof systems, for instance in privacy-preserving protocols. Designing lattice-based signatures that remain compatible with such proof systems is therefore crucial. A first attempt in this direction was the construction of Libert, Ling, Mouhartem et al. ([4]), which proposes a lattice-based signature scheme with efficient protocols. Although conceptually appealing, their construction suffers from extremely large proof sizes (several hundreds of megabytes) making it impractical. The paper studied in this report ([3]) revisits this problem and proposes a new lattice-based signature scheme specifically designed to interface smoothly with recent zero-knowledge argument systems, while reducing the size of the produced proofs. In the following, we analyse the main ideas behind this construction, highlight its improvements over previous work, and examine some of its technical components in more detail. We then try to apply this signature scheme to a "real life" scenario of e-voting by implementing it in python.

2 General Paper Analysis

The goal of the paper ([3]) is the design of lattice-based signatures that work well with zero-knowledge proof systems, a property known as "signatures with efficient protocols" (SEP). This is crucial because privacy-preserving authentication systems — like anonymous credentials, group signatures, e-cash and so on — require proving knowledge of a signature on a given message and because the systems deployed up to now revealed to be vulnerable to quantum computers.

One of the prior lattice-based SEP constructions is Libert et al. (Asiacrypt 2016) ([4]). In the paper, the authors found a way to make lattice-based signatures work with zero-knowledge proof systems. However, their proofs of knowledge reached hundreds of megabytes (the paper provides a fair and quantitative analysis of this limitation). All cited works appear in top-tier venues (Crypto, Eurocrypt, Asiacrypt, Journal of Cryptology), and the paper itself was published at Crypto 2023.

The paper studied in this report ([3]) aims to fill in the gap caused by the problem of these previous works. In order to do this, they generate the signature and commitment parameters together. This one change makes a difference: fewer witnesses, smaller SIS bounds, and much better parameters.

The methodology builds on well-established tools — Micciancio-Peikert trapdoors ([10]), Rényi divergence, and the Yang et al. ([11])/ LNP22 zero-knowledge frameworks — and is technically sound. The paper also identifies and corrects an error in the fast mode of Yang et al., which is a valuable independent contribution.

The results are amazing. The proof sizes go down from around 670 MB to below 650 KB in the module instantiation, a reduction of over three orders of magnitude.

The weaknesses are not fundamental in the sense that they arise from the methodology rather than from insurmountable barriers. The main practical weakness is the remaining proof size overhead relative to classical constructions, but it simply reflects the general current state of lattice-based zero-knowledge

proofs. It is plausible that further advances in proof systems will reduce these sizes further, and the paper's modular design allows to directly benefit from such future improvements.

The paper has attracted follow-up citations in the lattice cryptography literature, with works using SEP framework from this paper as a part of their considerations, consistent with the authors' stated goal of providing a versatile building block. These works generally affirm the paper's contributions and have not identified fundamental flaws. We can, however, note that some further efficiency improvements are highlighted and lead to more recent constructions (cf. 5).

This paper, presented at Crypto 2023, was the first one to provide a practical way to create anonymous credentials that can withstand quantum computers. It is likely to remain influential as long as lattice-based cryptography remains the dominant post-quantum paradigm.

3 Paper Contributions

3.1 Notations

In this report, we follow the notations used in [3]. We recall the most important notations here. We denote vectors by bold lowercase letters such as $\mathbf{a}$, and matrices by bold uppercase letters such as $\mathbf{A}$. The identity matrix of dimension n is written as $\mathbf{I}_n$. We use the notation $\mathbb{Z}_q$ for the ring $\mathbb{Z}/q\mathbb{Z}$, for a positive integer q. The uniform distribution over a given finite set S is denoted $U(S)$. The sampling $x \in S$ from a probability distribution P is denoted by $x \hookleftarrow P$. For a lattice $\mathcal{L}$ of rank n, we denote by $\mathcal{D}_{\mathcal{L},\sigma,\mathbf{c}}$ the discrete Gaussian distribution of support $\mathcal{L}$, width σ and center $\mathbf{c}$. If $\mathbf{c} = \mathbf{0}$, we omit it in the notations.

3.2 Large Sizes Problem

As mentioned in the introduction, the signature scheme from Libert et al. [4] suffers from large sized proofs (30 to 40 times larger). In this section, we study why this previous scheme yields such enormous sizes. In [4], the signature scheme relies on a commitment $\mathbf{c} = \mathbf{D_0 r} + \mathbf{D_1 m}$. To prove knowledge of a valid signature on a hidden message, the prover must demonstrate knowledge of the opening $(\mathbf{r}, \mathbf{m})$ such that $\mathbf{c}$ is well formed. More precisely, the verification algorithm operates on the binary decomposition of the commitment vector. In our case, $\mathbf{c} \in \mathbb{Z}_q^n$ so the protocol requires decomposing each coefficient c_i into $l = \lceil \log_2 q \rceil$ bits.
This design choice induces 2 main inefficiencies :

1. Witness expansion : The witness size for the ZKAoK scales linearly with the number of bits to be proven. Instead of proving knowledge of n elements in Z_q^n, the prover must prove knowledge of $n \cdot \log_2 q$ binary variables.

2. Constraint proliferation : To ensure the decomposition is valid, the ZK protocol must enforce the binary constraint for every bit. In frameworks based on quadratic constraints, this requires adding $\mathcal{O}(n \log q)$ quadratic equations to the proof system. Furthermore, the security reduction in [4] relies on embedding a trapdoor within the decomposition matrix. This introduces a norm growth factor $\propto \sqrt{n \log q}$ in the extracted solution, forcing the scheme to adopt overly conservative parameters to maintain security margins.

3.3 The Solution

To mitigate this, the authors optimize the interactions between the commitment scheme implicitly used by privacy-enhancing protocols and the signature scheme itself. Instead of generating the public parameters of these 2 components independently from one another, the authors propose a unified generation, using a common matrix $\mathbf{A}$. This indeed reduces the size of the proofs since the size of the public key is smaller and it allows the commitment randomness $\mathbf{r}$ to be algebraically absorbed into the signature vector $\mathbf{v}$, eliminating the need for a separate proof of opening.

4 Signature Scheme

The signature scheme from [4] on which the paper bases itself is a lattice-based signature with efficient protocols for multi-block messages. The paper studied ([3]) starts from this scheme and enhances it with the improvements discussed in the previous section.

The proposed construction is a lattice-based signature scheme with efficient protocols, designed to minimize the witness size required for subsequent zero-knowledge proofs. While it inherits the high-level goal of [4] (signing committed messages), its internal mechanics differ fundamentally to achieve the efficiency gains outlined in Section 3.3. The scheme relies on the hardness of the Short Integer Solution (SIS) problem over lattices and utilizes the trapdoor sampling framework of Micciancio and Peikert [10].

4.1 Key Generation and Parameters

The scheme operates over a modulus q and dimensions n, m. The key generation process (Algorithm 3.2 in [3]) produces a public key $\mathsf{pk} = (\mathbf{A}, \mathbf{B}, \mathbf{u})$ and a secret key $\mathsf{sk} = \mathbf{R}$.

- $\mathbf{A} \in \mathbb{Z}_q^{n \times m_1}$ is chosen uniformly at random. Crucially, in the context of signing hidden messages, $\mathbf{A}$ is derived from a public seed via a hash function (modeled as a Random Oracle) to ensure it is statistically random and independent of the trapdoor.
- $\mathbf{R} \in \{-1, 0, 1\}^{m_1 \times m_2}$ is a short random matrix serving as the *trapdoor*.
- $\mathbf{B} = \mathbf{AR} \pmod q$ is computed such that the relation between $\mathbf{A}$ and $\mathbf{B}$ is hidden by the shortness of $\mathbf{R}$.
- $\mathbf{u} \in \mathbb{Z}_q^n$ is a random syndrome defining the specific lattice coset for signatures.

The public parameters also include a gadget matrix $\mathbf{G} = \mathbf{I}_n \otimes \mathbf{g}$, where $\mathbf{g} \leftarrow [2^0 | \dots | 2^{\lceil \log_2 q \rceil - 1}] \in \mathbb{Z}_q^{1 \times \lceil \log_2 q \rceil}$ is a gadget vector, and a commitment key $\mathbf{D} \hookleftarrow U(\mathbb{Z}_q^{n \times m_3})$, which are used during the signing and verification phases.

4.2 Signing Committed Messages

The core innovation lies in the signing algorithm (Algorithm 3.3). To sign a binary message $\mathbf{m}$ hidden inside a commitment $\mathbf{c}$, the scheme employs a linear Ajtai commitment ([8]). the protocol proceeds as follows:

1. **Commitment:** The user computes a commitment to $\mathbf{m}$ using the public matrix $\mathbf{A}$:

$$\mathbf{c} = \mathbf{Ar} + \mathbf{Dm} \pmod q \tag{1}$$

   where $\mathbf{r}$ is a Gaussian randomness vector. Note that $\mathbf{A}$ is shared between the signature key and the commitment scheme.

2. **Preimage Sampling:** The signer chooses a tag $\tau \in \mathcal{T}$ and uses the trapdoor $\mathbf{R}$ to sample a short vector $\mathbf{v}'$ satisfying a modified linear equation:

$$[\mathbf{A} \mid \tau\mathbf{G} - \mathbf{B}]\mathbf{v}' = \mathbf{u} + \mathbf{c} \pmod q \tag{2}$$

   - To obtain $\mathbf{v}'$, the scheme calls the SampleD function, defined in [10], that, on input a trapdoor matrix $\mathbf{R} \in \mathbb{Z}^{m_1 \times n \lceil \log_2 q \rceil}$, a partial parity-check matrix $\mathbf{A} \in \mathbb{Z}_q^{n \times m_1}$, an invertible tag matrix $\mathbf{H} \in \mathbb{Z}_q^{n \times n}$ (in this case, defined as $\tau \mathbf{I}_n$), a syndrome $\mathbf{u} \in \mathbb{Z}_q^n$ and a standard deviation σ, outputs $\mathbf{v}$ that is statistically close to $\mathcal{D}_{\mathbb{Z}^{m_1 + n \lceil \log_2 q \rceil}, \sigma}$ conditioned on $[\mathbf{A} \mid \mathbf{HG} - \mathbf{AR}]\mathbf{v} = \mathbf{u} \bmod q$, with $\mathbf{G} = \mathbf{I}_n \otimes \mathbf{g}$ and $\mathbf{g} = [1 \dots 2^{\lceil \log_2 q \rceil - 1}]$ (cf. Lemma 2.6 in [3]).

   Substituting $\mathbf{c} = \mathbf{Ar} + \mathbf{Dm}$ and $\mathbf{B} = \mathbf{AR}$, the equation becomes:

$$\mathbf{A}\mathbf{v}_1' + (\tau\mathbf{G} - \mathbf{AR})\mathbf{v}_2' = \mathbf{u} + \mathbf{Ar} + \mathbf{Dm} \tag{3}$$

3. **Randomness Merging:** The final signature vector $\mathbf{v}$ is constructed by subtracting the commitment randomness $\mathbf{r}$ from the first part of $\mathbf{v}'$:

$$\mathbf{v} = \mathbf{v}' - \begin{bmatrix} \mathbf{r} \\ \mathbf{0} \end{bmatrix} \tag{4}$$

   The resulting signature is the pair $\sigma = (\tau, \mathbf{v})$.

4.3 Verification

Verification (Algorithm 3.4) checks the validity of the signature without knowing $\mathbf{r}$. The verifier computes:

$$[\mathbf{A} \mid \tau\mathbf{G} - \mathbf{B}]\mathbf{v} \stackrel{?}{=} \mathbf{u} + \mathbf{D}\mathbf{m} \pmod q \tag{5}$$

By substituting $\mathbf{v} = \mathbf{v}' - [\mathbf{r}^T \mid \mathbf{0}^T]^T$, the term $\mathbf{Ar}$ from the commitment cancels out with the $\mathbf{Ar}$ implicitly contained in $\mathbf{v}'$. This algebraic cancellation is the key mechanism: it allows the verifier to check the signature against the message $\mathbf{m}$ directly, without needing to verify the commitment opening $\mathbf{r}$ separately. Additionally, the verifier checks that $\|\mathbf{v}\|$ is sufficiently small to ensure security against lattice reduction attacks.

4.4 Security Considerations

The reuse of matrix $\mathbf{A}$ for both the signature trapdoor and the commitment introduces a potential vulnerability: if the signer could choose $\mathbf{A}$ maliciously, they might break the hiding property of the commitment. Indeed, $\mathbf{A}$ must look statistically random such that the term $\mathbf{Ar}$ looks like uniform noise in the commitment scheme. If the signer is allowed to choose $\mathbf{A}$ themselves, making $\mathbf{Ar}$ behave non-uniformly, it would not perfectly mask $\mathbf{Dm}$. It is then possible that $\mathbf{c}$ might leak statistical information about $\mathbf{m}$. To prevent this, the scheme requires $\mathbf{A}$ to be generated via a hash function (Random Oracle Model) or a trusted setup, ensuring $\mathbf{A}$ is statistically uniform and independent of the trapdoor $\mathbf{R}$. The security of the scheme is proven under the SIS assumption, i.e., that given a random matrix $\mathbf{A}$, finding a small vector $\mathbf{x} \in \mathbb{Z}_q^n \backslash \{\mathbf{0}\}$ such that $\mathbf{Ax} = \mathbf{0} \bmod q$ is hard. In the security analysis of the signature scheme, the authors distinguish between 2 types of forgeries.

- Type I forgery (cf. Lemma 3.2. in [3]) : An adversary produces a Type I forgery $(\tau^*, \mathbf{v}^*)$ if the tag τ^* does not collide with the tags of the signing queries. More informally, if the tag τ^* is a fresh, never-used tag.
- Type II forgery (cf. Lemma 3.3. in [3]) : An adversary produces a Type II forgery $(\tau^*, \mathbf{v}^*)$ if the tag τ^* is re-used from from some i^*-th signing query $(\tau^{(i^*)}, \mathbf{v}^{(i^*)})$, i.e., $\tau^* = \tau^{(i^*)}$.

Now we give an overview of the security proofs discussed above. Both reductions (Lemmas 3.2 and 3.3 of [3]) build an adversary $\mathcal{B}$ against SIS from a forger $\mathcal{A}$ against the signature: $\mathcal{B}$ embeds its SIS challenge in the public key given to $\mathcal{A}$, simulates the signing oracle, and extracts a short non-zero lattice vector from the forgery. The two cases differ in how the targeted signing query is simulated, which is where Rényi divergence enters.

4.4.1 Type I forgery.

On input a SIS challenge $[\mathbf{A} \mid u]$, $\mathcal{B}$ generates the deterministic tags $\tau^{(1)}, \ldots, \tau^{(Q)}$ in advance, picks a guess $\overline{\tau} \hookleftarrow \mathcal{T} \setminus \{\tau^{(i)}\}$ on the forgery tag, and samples $\mathbf{U} \hookleftarrow \{-1, 0, 1\}^{m_1 \times m_3}$, $\mathbf{R} \hookleftarrow \{-1, 0, 1\}^{m_1 \times m_2}$. It sets $\mathbf{D} = \mathbf{AU}$ and $\mathbf{B} = \mathbf{AR} + \overline{\tau}\mathbf{G}$, so that

$$\mathbf{A}_\tau = [\mathbf{A} \mid (\tau - \overline{\tau})\mathbf{G} - \mathbf{AR}]. \tag{6}$$

For $\tau^{(i)} \neq \overline{\tau}$, the difference is invertible in $\mathbb{Z}_q$, so $\mathbf{R}$ remains a G-trapdoor and $\mathcal{B}$ answers all queries via `SampleD`. The Leftover Hash Lemma guarantees that the simulated view is within negligible statistical distance of the real one.

If $\tau^* = \overline{\tau}$ (probability $1/(|\mathcal{T}| - Q)$), the $\tau\mathbf{G}$ term in (6) vanishes and the forgery yields

$$w = \begin{bmatrix} [\mathbf{I}_{m_1} \mid -\mathbf{R}]v^* - \mathbf{U}m^* \\ -1 \end{bmatrix} \in L_q^\perp([\mathbf{A} \mid u]).$$

The trailing -1 guarantees $w \neq 0$ deterministically. The norm bounds β_∞, β_2 follow from the verification check on v^*, $\|\mathbf{R}\|_2 \leq \sqrt{m_1} + \sqrt{m_2} + t$ (Lemma 2.4 of [3]), and the sub-exponential tail bound on $\|\mathbf{U}m^*\|_2$ (Lemma 2.5 of [3]), the latter being the technical improvement that avoids the suboptimal $\sqrt{m_3}$ factor of [4].

4.4.2 Type II forgery and Rényi divergence.

When $\tau^* = \tau^{(i^+)}$ for some guessed query index i^+, the cancellation of $\mathbf{G}$ now happens *at* i^+, so $\mathcal{B}$ cannot use `SampleD` for that query. It instead samples $v \hookleftarrow D_{\mathbb{Z}^m,\sigma}$, $r_0 \hookleftarrow D_{\mathbb{Z}^{m_1},\sigma_2}$, and programs $u := \mathbf{A}_{\tau^{(i^+)}}(v - [r_0^T \mid 0]^T) \bmod q$. On the i^+-th query, it returns $v^{(i^+)} = v - [r_0^T - (\mathbf{U}m^{(i^+)})^T \mid 0]^T$.

The gap. Whereas the real $v_1^{(i^+)} \sim D_{\mathbb{Z}^{m_1},\sigma_1}$ is centred at 0, the simulated one is shifted to $D_{\mathbb{Z}^{m_1},\sigma_1,\,z^+}$ with $z^+ = \mathbf{U}m^{(i^+)}$. The statistical distance scales as $\Theta(\|z^+\|_2/\sigma_1)$, requiring $\sigma_1 \gg 2^\lambda \|z^+\|_2$ for negligibility, which would blow up all parameters. Rényi divergence (Lemma B.2 of [3]) gives instead

$$\mathrm{RD}_\alpha\big(D_{\mathbb{Z}^{m_1},\sigma_1} \,\big\|\, D_{\mathbb{Z}^{m_1},\sigma_1,z^+}\big) \le \exp(\alpha\pi\|z^+\|_2^2/\sigma_1^2),$$

which is $e^{\alpha\pi}$ as soon as $\sigma_1 \ge \|z^+\|_2$ — exactly the condition imposed by Algorithm 3.1.

Probability preservation and optimisation. For any event E, Lemma B.1 of [3] gives $P(E)^{\alpha/(\alpha-1)} \le \mathrm{RD}_\alpha(P\|Q)\cdot Q(E)$, so an adversary with real advantage δ has simulated advantage at least $e^{-\alpha\pi}\delta^{\alpha/(\alpha-1)}$. The optimal order $\alpha^* = 1+\sqrt{\log_2(1/\delta)/(\pi\log_2 e)}$ depends on δ, but neither $\mathcal{B}$ nor the scheme parameters do — α is purely an analytical knob. The final advantage is $\mathrm{Adv}[\mathcal{B}] \gtrsim \delta^{\alpha^*/(\alpha^*-1)}e^{-\alpha^*\pi}/Q$.

Non-triviality of w is not automatic here: $w = 0$ would force $\mathcal{A}$ to predict a column of $\mathbf{U}$ from $(\mathbf{A}, \mathbf{A}u')$, which has conditional min-entropy at least $m_1\log_2 3 - n\log_2 q \ge \omega(\log_2\lambda)$ by the Leftover Hash Lemma, hence negligible.

4.4.3 Lemma 4.1: fix to the [11] fast mode.

[11] claims that with binary $\mathbf{H}$, proving $\mathbf{H}w \in [0, nB]^k$ implies $w \in [0, nB]^n$ w.h.p. The counter-example $w = [-1, 1, \ldots, 1]^T$ defeats this: each binary row satisfies the bound with probability $1 - 2^{-n}$, so the implication fails. The fix in [3] switches to ternary $\mathbf{H} \sim U(\{-1, 0, 1\}^{k\times n})$ with $k = \lambda/\log_2(9/5)$, for which Lemma 4.1 of [3] establishes that $\|w \bmod q\|_\infty > 2B$ implies $\Pr[\|\mathbf{H}w \bmod q\|_\infty \le B] \le (5/9)^k$, via a combinatorial argument on equivalence classes in $\mathbb{Z}_3^{n-1}/\langle\mathbf{1}\rangle$. This both fixes the soundness gap (from nB to $2B$) and tightens parameters.

4.4.4 Module setting (Lemmas 3.5 and 3.6).

The module reductions follow the same skeleton, with four ring-specific substitutions: (i) tag-difference invertibility via [[7], Cor. 1.2], which constrains $q \equiv 2k+1 \pmod{4k}$ and $\|\tau_1 - \tau_2\|_\infty \le 2$; (ii) spectral norm of nega-circulant block matrices, bounded via the n complex embeddings and a union bound [[2], Lem. 2.3]; (iii) ring Leftover Hash Lemma [[2], Lem. 2.8]; (iv) regularity of $\mathbf{A}v \bmod q\mathcal{R}$ via [[6], Thm. 7.4]. The Rényi argument is unchanged, since Gaussians over $\mathcal{R}$ are defined through the coefficient embedding into $\mathbb{Z}^n$.

4.5 Cyclotomic Ring

A major step in the paper is the move from standard lattices to module lattices. Instead of working only with integers, the construction is transferred to the ring

$$R = \mathbb{Z}[X]/\langle X^n + 1\rangle,$$

where n is a power of 2. In the paper, this ring is the $2n$-th cyclotomic ring. The authors also do modulo q, using

$$R_q = \mathbb{Z}_q[X]/\langle X^n + 1\rangle.$$

This setting is attractive in post-quantum lattice cryptography because the additional structure leads to more efficient computations and more compact keys. The signature equation keeps the same general shape as in the standard lattice setting, but it is now written over the ring:

$$[\mathbf{A} \mid \tau\mathbf{G} - \mathbf{B}]\mathbf{v} = \mathbf{u} + \mathbf{D}\mathbf{m} \pmod{qR},$$

with the extra condition that $\mathbf{v}$ has to remain short. Here, the objects are no longer ordinary integer vectors but vectors of polynomials. This means that some notions have to be defined differently. Like, a binary polynomial means that all its coefficients are in $\{0, 1\}$, while a short element means that its coefficients remain small.

Proving knowledge of a valid signature requires proving four things at the same time: that the tag τ belongs to the chosen tag space; that $\mathbf{v}$ is short; that $\mathbf{m}$ is a vector of binary polynomials; and that the verification equation is satisfied. The paper explains that these conditions are not treated separately, but are all handled inside the LNP22 framework [5]. This is important because the efficiency of the scheme depends not only on the signature, but also on how well it fits the proof system used. This module version is the most practical one in the paper. It is the combination of the ring structure and the LNP22 proof framework that makes the proof of message-signature possession drop below 650 KB. Using this construction as a building block, the authors then build an anonymous credential system.

5 Comparison With More Recent Constructions

The constructions presented in this paper are now superseded by the constructions in the more recent paper [1]. Indeed, the constructions of [3] representing a step toward a realistically implementable signature scheme with efficient protocols, still remains impractical due to several structural and inefficiencies in the proofs. In particular, achieving adaptive security requires the inclusion of a second MP trapdoor in the public key, this significantly grows matrix dimensions, signature sizes, and zero-knowledge witnesses. Moreover, the scheme in [3] relies mainly on statistical security arguments, which impose conservative parameter choices resulting in large proofs of possession. Additional inefficiencies are coming from the use of spherical perturbation in preimage sampling and the injection of signer-side randomness into the syndrome. These result in increased norm bounds and further degraded performance. The work [1] addresses these limitations. It introduces a partial trapdoor switching technique that removes the need for a full auxiliary trapdoor while preserving a proof under standard M-SIS and M-LWE assumptions, yielding much smaller proofs and signatures. The authors also replace spherical sampling with an elliptic sampler, tighten probabilistic and norm analyses, and eliminate signer-side randomness via a rejection-sampling-based security argument. All these improvements allow for reduced parameter sizes, yielding in much smaller proofs of possession of about $80KB$, enabling, for the first time, a practical lattice-based instantation of anonymous credentials under standard assumptions.

6 Application to E-Voting

6.1 System Architecture

In the following section we use the following syntax (cf. Section C.1.1. in [3]), an anonymous credentials is composed of two algorithms and two interactive protocols. The OKeyGen takes as input public parameters (those generated by Setup) and outputs the organization's key pair $(\mathsf{opk}, \mathsf{osk})$. Similarly, the UKeyGen takes the public parameters and outputs the user's key pair $(\mathsf{upk}, \mathsf{usk})$. The $\mathsf{Issue}_{O,U}$ protocol involves an organization O and a user U. The user either obtains a credential cred on its attributes $(\mathbf{m}_i)_{i\in[l]}$ or $\perp$ if the protocol failed, on the other hand, the organization just gets notified of the success (or not) of the execution. Finally, the $\mathsf{Show}_{U,V}$ protocol involves a user U and a verifier V. It outputs $b = 1$ to V if the credential cred is valid and $b = 0$ otherwise. We map the anonymous credentials construction of Section 5.3 of [3] to an e-voting system. The mapping is shown in Figure 1 and Table 1.

Table 1: Mapping from Anonymous Credentials to E-Voting

Paper (Section 5.3)	**E-Voting System**	**Algorithm**
Organization	Election Authority	—
OKeyGen	Election key generation	Alg. 5.3 → Alg. 3.2
UKeyGen	Voter registration	Alg. 5.4
Issue	Credential issuance	Alg. 5.5 (oblivious signing)
Show	Vote casting	Alg. 5.6 (ZK proof)
Tag $\tau = F(\mathsf{st})$	Anti-double-vote serial	Stateful, injective F
Bulletin board	Public vote record	Tags + proofs

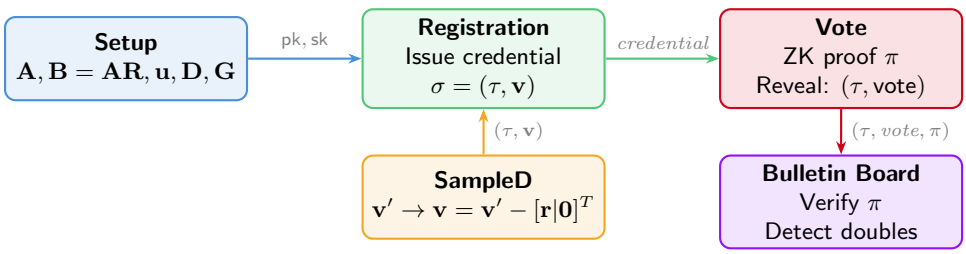

Figure 1: E-voting pipeline based on the lattice SEP construction.

6.2 Protocol Description

The e-voting protocol proceeds in four phases.

Phase 1: Election Setup. The election authority runs KeyGen (Algorithm 3.2) to generate its signing keys $(\mathsf{pk}, \mathsf{sk})$. The public parameters pp and the public key $\mathsf{pk} = (\mathbf{A}, \mathbf{B}, \mathbf{u})$ are published. The secret key $\mathsf{sk} = \mathbf{R}$ is stored securely.

Phase 2: Voter Registration. Each voter generates a secret key $\mathbf{s} \leftarrow_{\$} \{0,1\}^{m_s}$ via $\mathsf{UKeyGen}$ (Algorithm 5.4). The voter then interacts with the authority through the Issue protocol (Algorithm 5.5): the authority signs the message $\mathbf{m}_e = [\mathbf{s} \| \text{attributes}]$ via oblivious signing, producing a credential $\sigma = (\tau, \mathbf{v})$. The tag $\tau = F(\mathsf{st})$ is unique per voter, where F is an injective function and st is a counter incremented after each issuance.

Phase 3: Vote Casting. To cast a vote, the voter runs the Show protocol (Algorithm 5.6): they produce a zero-knowledge proof π demonstrating that they possess a valid credential $(\tau, \mathbf{v})$ on some message $\mathbf{m}_e$ satisfying $\mathsf{Verify}(\mathsf{pk}, \mathbf{m}_e, (\tau, \mathbf{v})) = 1$. The proof reveals *only* the tag τ and the vote — the voter's identity, secret key, and other attributes remain hidden.

Phase 4: Verification and Tallying. The bulletin board accepts a ballot (τ, vote, π) if and only if (i) the proof π verifies, and (ii) the tag τ has not been used before. If a voter attempts to vote twice, their tag will already appear on the board and the second ballot is rejected.

6.3 Security Properties

The system achieves the following security properties, directly inherited from the anonymous credentials construction. Let us first recall the definition of the Short Integer Solution (cf. Definition 2.3. in [3]). Let n, m, q be positive integers, and $\beta_2 \geq \beta_\infty \geq 1$. The Short Integer Solution problem $\mathrm{SIS}^{\infty,2}_{n,m,q,\beta_\infty,\beta_2}$ asks to find $\mathbf{x} \in \mathcal{L}^{\perp}_q(\mathbf{A}) \backslash \{\mathbf{0}\}$ given $\mathbf{A} \hookleftarrow U(\mathbb{Z}_q^{n \times m})$ such that $\|\mathbf{x}\|_\infty \leq \beta_\infty$ and $\|\mathbf{x}\|_2 \leq \beta_2$.

1. **Anonymity.** The bulletin board reveals only (τ, vote). The link between a voter's identity and their tag is protected by the zero-knowledge property of the proof system (LNP22 framework [5]).

2. **Unforgeability.** Creating a valid credential without the authority's secret key requires solving the SIS problem, which is believed to be hard even for quantum computers. The EUF-CMA security of the signature reduces to $\mathrm{SIS}^{\infty,2}_{n,m_1,q,\beta}$ via the proofs in Lemmas 3.2 and 3.3 of [3].

3. **Double-vote prevention.** Each voter receives a unique tag τ. Since tags are revealed on the bulletin board, any duplicate is immediately detected.

4. **Post-quantum security.** The underlying Module-SIS and Module-LWE problems are not known to be solvable in polynomial time by quantum algorithms.

7 Implementation

7.1 Code Structure

Our implementation is organized into five Python modules, with a clear separation between the lattice primitives and the e-voting application layer:

Table 2: Source code organization

Module	Description
`lattice_sampler.py`	`SampleD` algorithm with corrected gadget decomposition, `gadget_solve` base-2 decomposition
`signature.py`	`LatticeSignatureScheme` class implementing Algorithms 3.1–3.4
`evoting.py`	`ElectionAuthority`, `BulletinBoard`, `Voter` classes mapping Section 5.3
`parameters.py`	`ParameterAnalyzer` reproducing Table H.4
`main.py`	Entry point running the full simulation

7.2 Toy vs. Real Parameters

As the implementation is done in a Python simulation, some choices needed to be made for computational feasibility without changing the underlying structure. The choice here is to reduce by half the security margin. As the SIS dimension determines hardness, the paper uses $n = 128$, targeting 128-bits post-quantum security. In our simulation, $n = 64$ is used to halve matrix sizes. As the matrix is a power of two, the lattice structure is still valid. The commitment randomness dimension (size of matrix A) $m1$ is set to 128 (657 in the paper) to greatly reduce computing time to make 99999 voters registration in reasonable time. m_2 (trapdoor dimension) $m_2 = n \cdot \lceil \log_2 q \rceil = 64 \cdot 12 = 768$ is derived automatically with $q = 3329$ (Kyber/ML-KEM prime chosen because it is small and supports efficient arithmetic). As for the gaussian widths, the paper's widths are derived from a full security proof. The toy widths are smaller because the dimensions are smaller. σ_2 (commitment randomness r) which binds the message m in to the commitment needs to be short enough to hide the message. $\sigma_2 = 500$ is chosen small enough that the commitment is stastically hiding while keeping r short. $\sigma = 2000$ as it samples the perturbation ρ inside `SampleD`. It must be larger than σ_2 so it dominates and hides the trapdoor R. σ_1 is derived accordingly ($= \sqrt{\sigma^2 + \sigma_2^2} \approx 2061$). It's not sampled, just the expected width of v_1 used for verification bound check.

8 Parameter Analysis

8.1 Comparison with Prior Work

Table 3 reproduces Table 1.1 from [3], comparing the signature scheme with the only prior lattice-based SEP construction [4].

Table 3: Efficiency comparison (128-bit quantum security, $\lambda = 128$)

Metric	**LLM$^+$16 [4]**	**This paper (std.)**	**This paper (module)**
$\lvert\mathsf{pk}\rvert$	707×10^4 MB	296×10 MB	**7.8 MB**
$\lvert\mathsf{sk}\rvert$	372×10^2 MB	229×10 MB	**8.9 MB**
$\lvert\sigma\rvert$	2139×10^2 KB	418 KB	**273 KB**
$\lvert\pi\rvert$ (ZK proof)	2671×10^3 KB	177×10^2 KB	**639 KB**

The module instantiation achieves improvements of approximately 900× for the public key, 4 000× for the secret key, 780× for the signature, and 4 000× for the ZK proof, compared to [4].

8.2 E-Voting Scalability

We evaluate the practical feasibility of deploying the scheme for elections of various sizes. The dominant cost is the storage of ZK proofs on the bulletin board, at 724 KB per voter.

Table 4: Bulletin board storage requirements

Number of voters	**Total storage**
1 000	0.7 GB
10 000	6.9 GB
100 000	69 GB
1 000 000	691 GB

For comparison, the pairing-based Helios system [9] produces proofs of approximately 2–5 KB, yielding an overhead factor of roughly 240×. However, Helios is not post-quantum secure. The tag space $T_w = \{\tau \in S_{\text{bin}} : \|\tau\|_2 = \sqrt{w}\}$ with $w = 6$ and $n = 128$ has cardinality $\binom{128}{6} \approx 5.4 \times 10^9$, sufficient for any national election.

9 Experimental Results

We ran our simulation with 10 voters and 3 candidates using toy parameters. The results confirm the correctness and security properties of the implementation.

Table 5: Simulation results (toy parameters)

Metric	**Result**
Signatures issued	10/10 valid
Votes cast and verified	10/10 accepted
Double-vote attempts	1 detected and rejected
Forgery attempts	1 correctly rejected
Signing time (avg.)	< 1 ms
Verification time (avg.)	≈ 2 ms
Signature size (toy)	7.0 KB
Public key size (toy)	448.5 KB

All 10 signatures passed the verification equation $\mathbf{A}_\tau \mathbf{v} = \mathbf{u} + \mathbf{D}\mathbf{m} \bmod q$ and the norm bounds $\|\mathbf{v}_1\|_\infty \leq \sigma_1 \log_2 m_1$ and $\|\mathbf{v}_2\|_\infty \leq 1$ (since $\mathbf{v}_2$ contains binary decomposition digits). The double-vote attempt was rejected because the tag $\tau = 1$ had already been recorded on the bulletin board. The forgery attempt was rejected because a random vector $\mathbf{v}$ has negligible probability of satisfying the verification equation.

10 Work Organization

First, the members of the group studied the paper on their own, in order to have a good understanding of the content. A meeting was then held in order to exchange any questions and to give a better comprehension of the article to every member of the group. As for the work distribution, 2 groups have been made, one group (3 members) focusing on the theoretical part and one group (2 members) focusing on the practical part. Even though the groups were studying different branches of the project, it is still very important to understand everything that is being done, therefore, meetings have been organized to share what the students did with each other. The main difficulties encountered were the understanding of the `SampleD` algorithm at a level of detail sufficient to implement and debug it. In particular, identifying the gadget decomposition bug (base-2 decomposition vs. direct modular reduction) required careful comparison between the paper's description of $\mathbf{G} = \mathbf{I}_n \otimes \mathbf{g}$ and the actual code behavior. Another challenge was mapping the abstract anonymous credentials framework (Section 5.3 of [3]) to a concrete e-voting scenario, which required design decisions not covered by the paper (e.g., how to encode voter attributes, what constitutes a ballot on the bulletin board). Another difficulty was to keep this report under 10 pages, some part have been cut in order to respect this constraint.

11 Conclusion

In conclusion, the paper makes an important contribution to lattice-based cryptography by addressing a challenge that is both technical and practical: how to design a post-quantum signature scheme that remains efficient in privacy-preserving applications. Rather than making a simple optimization, the authors redesign the interaction between the signature and the commitment mechanism, significantly reducing the size of the zero-knowledge proofs required to prove possession of a valid signature on a hidden message. What makes the construction particularly interesting is that these efficiency gains do not come at the cost of security. The scheme still relies on standard lattice assumptions, like SIS, while using trapdoor sampling and careful parameter design. The paper shows that efficiency in modern cryptography depends not only on the security of a primitive itself, but also on how well it interacts with other components such as commitments and zero-knowledge proofs. More generally, it shows that lattice-based constructions can be made much more practical for real-world privacy applications. On the implementation side, we have implemented the signature scheme (Algorithms 3.1–3.4), and built a complete e-voting simulation demonstrating the anonymous credentials construction.

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